ensia The National School of Artificial Intelligence المدرسة الوطنية العليا للذكاء الاصطناعي

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Advancing Legal Reasoning in *Algerian Law*

RAG · KNOWLEDGE GRAPHS · ARGUMENT MINING → *Citation-faithful legal answers*

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01 The Problem

WHY WHY LEGAL QA IS HARD

1.1 Fluent is not lawful

A legal answer cannot be judged by fluency or plausibility. To be valid it must be **grounded in identifiable sources**, cite the **exact governing articles**, stay inside the **Algerian jurisdiction**, and read each provision **as amended** at the relevant date — and where the law is silent, the only correct answer is a **refusal**. A fluent language model guarantees none of this.

SOURCED · ARTICLE-CITED · JURISDICTION-BOUND · AMENDMENT-AWARE · *OR REFUSED*

1.2 What legal questions demand

بيّن الأساس القانوني لمبدأ خضوع القاضي للنص التشريعي، وحدود سلطته عند غياب النص…

— *demands* **exact articles** — *Civil Code, arts. 1–2*

كيف تغيّرت شروط الخلع في قانون الأسرة بعد تعديل 2005؟

— *demands the law* **as amended** — *Ord. 05-02 → Family Code, art. 54*

ABSTAIN

هل يحق لصاحب العمل في الجزائر فصل العامل بصفة تامة دون إبداء أسباب (at-will)؟

— *a* **foreign-doctrine trap** — *Algerian law is silent; the system must* **abstain**

verbatim from AlgerianLegalBench (244 questions)

1.3 Where today's LLMs fail

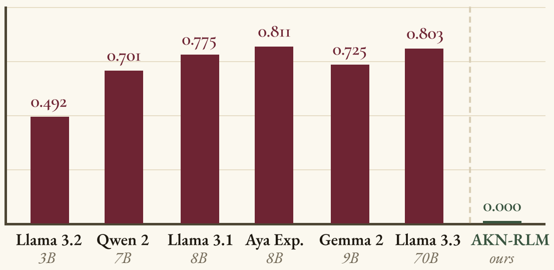


Fig. 1. Hallucinated-citation rate: the share of cited articles that *do not exist*, per direct-LLM baseline (244 questions). Scale does not help — the 70B model fabricates 4 of 5 citations. Gemini 2.5 reaches 0.008 only by declining to cite at all, and six of seven baselines import foreign law on >12 of 40 trap questions. **AKN-RLM: 0.000.**

1.4 …and Algerian law raises the bar

- **Arabic-first.** 232 of 244 benchmark questions are in Arabic
- **Layered in time.** 1963–2025; amendments rewrite articles in place
- **Deeply structured.** code → chapter → section → article
- **Low-resource.** barely covered by existing legal-AI corpora

1.5 Objectives

- ✓ **Structure the law** — *171 docs → Akoma Ntoso XML + legal KG*
- ✓ **Benchmark it** — *AlgerianLegalBench: 244 expert questions*
- ✓ **Reason on evidence** — *typed retrieval + bounded recursion*
- ✓ **Verify or abstain** — *triple faithfulness gate*

1.6 Significance & impact

1. **First citation-faithful QA system for Algerian law** — every claim resolves to a registered article, or the system abstains.
2. **Reusable infrastructure for Algerian legal AI**
   - **AKN corpus + knowledge graph** — 171 structured documents, 1963–2025, article-level registry
   - **AlgerianLegalBench** — 244 expert-annotated questions, κ = 0.829
3. **A recipe for low-resource legal AI** — evidence control beats parameter scale.

02 The Solution

HOW STRUCTURED CORPUS → BENCHMARK → AKN-RLM

I Structured legal corpus

the law, made machine-readable

What is Akoma Ntoso? The OASIS LegalDocML standard — an XML vocabulary for legal texts in which every law, article, amendment and cross-reference becomes an **addressable node** under a canonical FRBR URI: the article registry that every citation must resolve against.

WHAT WE BUILT

171	source files 189 MB
45	laws & codes → AKN XML
46	RDF graphs → one legal KG
1963–2025	coverage, amendments in place

```
<!-- Civil Code — Ordinance 75-58, as amended -->
<act name="code_civil">
  <FRBRWork value="/akn/dz/act/1975-09-26/75-58"/>
  <FRBRExpression value="…/ara@2005-06-20"/>
  <lifecycle>
    <eventRef date="2005-06-20" type="amendment"
              source="#law-05-10"/>
  </lifecycle>
  <article eId="art_124">
    <num>المادة 124</num>
    <content>كل فعل أيا كان يرتكبه الشخص بخطئه…</content>
    <ref href="#art_127"/>
  </article>
</act>
```

Validated in **multiple manual passes** by two law students and a supervising senior legal expert — the structure is derived from the sources, never invented.

II AlgerianLegalBench

the evaluation layer

244 expert-annotated questions across **23** legal categories — **177 answerable**, **67 expecting abstention** · 232 Arabic / 12 French · each carries expected articles, a reference answer and a **four-step reasoning chain** · double-annotated, **κ = 0.829**.

QUERY TYPES

Rule application	66
Exact article	59
Unanswerable	40
Multi-hop	26
Layman	17
Long-context	17
Conceptual	12
Temporal	7

DIFFICULTY

Easy	Medium	Hard
23%	35%	42%

40 jurisdictional-infection traps — *questions seeded with French, US, Egyptian, Tunisian & Gulf canary doctrine the system must refuse to import.*

III AKN-RLM

the answer pipeline — Fig. 3.2

User query — arabic · french · darja

↓

Normalize — NFC · hamza · diacritics | **Classify intent** — gemma 4-31B · 8 types | **Decompose** — qwen3 · MH / LC only

↓

Doc-router — alias + numeric-id + BM25 aggregation → top-3 candidate codes

↓

Typed-handler dispatch (1 of 8) — EA · RA · MH · TF · CD · UN · Layman · LC

↓

recursion (max depth 3)

BM25 — legal-id tokenizer | **Dense retrieval** — e5-small · FAISS | **HyDE** — qwen3 · off for TF, CD

------ *RRF fusion across active channels* ------

Amendment chain — sparql · CD / MH | **Concept-at-date** — sparql · TF | **Article registry** — corpus-boundary check

------ *765k triples · 8.9k articles* ------

Verifier — qwen3-30B · per-candidate | **Pervasive Toulmin ADU** — gemma · claim / ground / warrant | **Gap-probe recursion** — gpt-oss-120b · max depth 3

↓

Summarizer (qwen3-30B) — arabic answer over verified citations

↓

corrective retry (once)

Citation existence — akn registry check | **Jurisdiction gate** — canary detector · 40 anchors | **Faithfulness NLI** — mDeBERTa · per-citation ≥ 0.55

↓

Answer + citations — 4-step toulmin chain · trajectory · abstention path

03 Results

PROOF FROZEN 244-QUESTION RUN

0.305 Citation F1 — *article-level · locked*

✓ 0.000 — HCR · JIR

✓ 0.703 — Abstention F1

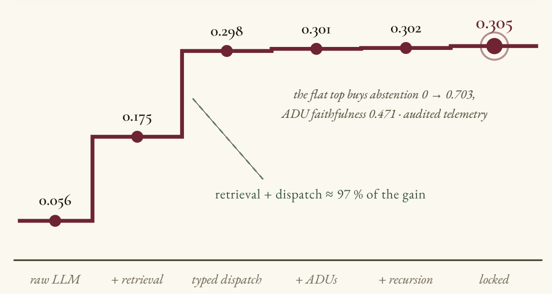
1.64×	1.74×	2.9×
vs best direct LLM	*vs minimal RAG*	*vs determ. retrieval*

+0.199 over the strongest same-corpus baseline, $p_{Holm} < 0.0001$ · document-level F1 0.546

`rlm_dispatched_full_phase_e_final · 2026-05-14 · depth ≤ 3 · NLI ≥ 0.55 · one retry → abstain`

The ablation ladder

CITATION F1 BY PHASE



Each phase is an ablation in disguise — **every layer pays for itself**, *though not always in F1.*

Citation F1 vs baselines

HIGHER = BETTER

Direct LLM **raw**	0.056
Determ. **hybrid+rerank**	0.105
Minimal RAG **top-5**	0.175
Best LLM **Gemini 2.5 Flash**	0.186
AKN-RLM *locked*	0.305

Jurisdictional infection (JIR)

LOWER = SAFER

AKN-RLM *triple gate*	0.000
Gemini 2.5 **15 / 40 traps**	0.375
Qwen 2 **27 / 40 traps**	0.675

Only AKN-RLM holds **0.000** hallucinated-citation & foreign-law rates. Reliability tracks **legal structure, not scale** — the largest baselines hallucinate most.

Reading the results

INTERPRETATION

i **The gain is architectural** — *identical corpus, registry & indices as every tier-2 baseline — the layers, not the scale, earn the F1*

ii **Strictness has a price** — *the gate over-rejects layman queries; wider evidence pools dilute citation discipline*

iii **Refusal works** — *38 / 40 infection traps correctly refused — every baseline invents an answer*

⚠ LIMITS

- Retrieval ceiling: R@10 = 0.258
- French under-served
- Abstention precision 0.654
- Metrics are proxies

NEXT

- Corpus-tuned embedder
- French ↔ Arabic retrieval
- Graph-retrieval redesign
- Expert legal review

Structured sources + Controlled retrieval + Recursive reasoning + Citation verification =

THE TAKEAWAY

Trustworthy Legal AI

an *evidence-controlled system*, not an unconstrained generator.